# BURHAN ARSHAD

**AI & Machine Learning Developer | Computer Science Student**

Gujranwala, Pakistan | burhanarshad707@gmail.com

linkedin.com/in/burhan-arshad | github.com/burhan-arshad | burhan.cypherai.tech | Upwork Profile

## PROFESSIONAL SUMMARY

Computer Science student (CGPA 3.99/4.00) at the University of Central Punjab with a focus on machine learning, deep learning, NLP and computer vision. Builds and deploys end-to-end intelligent applications, from model development and evaluation to FastAPI services and cloud-hosted demos. Additional freelance experience delivering WordPress and Shopify websites for international and local clients.

## EDUCATION

**University of Central Punjab (UCP)** Pakistan

Bachelor of Science in Computer Science (currently 4th Semester) | **CGPA: 3.99 / 4.00**

## TECHNICAL SKILLS

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|---|---|
| **Languages** | Python, C#, C++, JavaScript, SQL, HTML, CSS |
| **ML & Deep Learning** | Scikit-learn, TensorFlow, Keras, OpenCV, CNNs, LSTM, NumPy, Pandas, Matplotlib, Seaborn |
| **Generative AI & NLP** | LangChain, ChromaDB, Sentence-Transformers, Hugging Face, Groq, RAG, TF-IDF |
| **Backend & Deployment** | FastAPI, REST APIs, WebSockets, Streamlit, PostgreSQL, Supabase, Vercel, Render |
| **Web & eCommerce** | WordPress, Shopify (Liquid), Astro, Tailwind CSS, GSAP, Framer Motion |
| **Tools** | Git, GitHub, VS Code, Jupyter Notebook, Unity, Blender |

## SELECTED PROJECTS

**Real-Time ASL Sign Language Recognition** Live Demo • GitHub

*Python, TensorFlow/Keras, CNN, FastAPI, WebSockets*

- Trained a custom CNN on 87,000 images across 29 classes, achieving 99.19% validation accuracy.
- Delivered live webcam predictions through a FastAPI WebSocket pipeline deployed on Render.

**RAG Document Assistant** Live Demo • GitHub

*Python, LangChain, ChromaDB, Sentence-Transformers, Groq, Streamlit*

- Built a RAG app for question answering over uploaded PDF/TXT files using local embeddings, ChromaDB and MMR retrieval.
- Integrated a Groq-hosted LLM for answers and surfaced retrieved source chunks for traceability.

**Chest X-Ray Pneumonia Detection** Live Demo • GitHub

*Python, TensorFlow/Keras, CNN, Streamlit*

- Developed a CNN classifier for normal vs. pneumonia X-rays, optimized for pneumonia-class recall (94%) to reduce missed positive cases.

**BTC Directional Signal: LSTM with Backtested Evaluation** Live Demo • GitHub

*Python, TensorFlow/Keras, LSTM, Pandas, Streamlit*

- Modeled short-term price direction from 100,000 hourly candles and 15 engineered technical indicators.
- Evaluated with out-of-sample backtesting (transaction costs, Sharpe ratio, max drawdown, win rate) instead of accuracy alone.

**Cloud-Based Multi-Tenant SaaS Inventory System** Live Demo • GitHub

*PostgreSQL, Supabase, SQL, Realtime/WebSockets, TypeScript*

- Designed a multi-tenant platform with tenant isolation, relational schema, authentication and real-time data sync; deployed on Vercel.

## EXPERIENCE

**Freelance Web Developer** | Upwork & Independent Remote

- Developed a WordPress website for Pearl Essence Interiors, a UAE-based interior design and fit-out company, presenting services, projects and brand identity.
- Built a responsive Shopify store for KADME, a Pakistani footwear brand, focused on product presentation, collection structure and conversion-oriented design.

## LEADERSHIP

**Coding Club Coordinator** | University of Central Punjab

- Coordinate coding-related sessions and presentations for the university Coding Club.